🧠 NeuroLink 2.0

Personal Typing Performance Analytics & Ghost Prediction Engine

NeuroLink 2.0 is a Python-based behavioural analytics engine that analyses large-scale typing history data to model personal typing performance and predict how long a user would take to type any given text.

Built on **30,000+ real typing sessions**, the project combines statistical analysis, behavioural modelling, and simulation to create a personalized **"Ghost" typing predictor** — a digital reflection of your own typing behaviour.

⚙️ What the Project Does

NeuroLink 2.0 performs three core tasks:

1. Load & preprocess typing history

- Reads a CSV file containing typing session data
- Sorts sessions chronologically
- Uses recent performance to reflect current skill

2. Extract personal performance metrics

From historical data, the engine computes:

- **Recent average WPM** (last 500 sessions)
- **Error penalty**: how much typing speed drops as errors increase (via linear regression)

These metrics form the user's personal *typing profile*.

3. Ghost typing simulation

Given any input text, NeuroLink:

- Estimates base typing time from personal WPM
- Simulates realistic typing errors
- Applies an individualized time penalty
- Outputs a **predicted completion time**

This simulated typist is called your **Ghost**.

👻 The Ghost Simulator

The Ghost represents:

*“How long **you**, not an average person, would take to type this text.”*

Prediction Logic (Simplified)

1. Convert WPM → characters per second
2. Estimate base typing time from text length
3. Simulate a small number of typing errors
4. Apply a personal error penalty
5. Output total predicted time (seconds)

This makes predictions:

- Personalized
- Consistent
- Comparable across sessions

✍️ Practice Sentence Generator

NeuroLink includes a built-in **practice text generator**, allowing you to:

- Generate a new sentence on every run
- Practice typing against your Ghost
- Avoid memorization bias

This turns the project into an **interactive training tool**, not just an analysis script.